

Homework 2

Diya Jain U81673988

1. a. $4 \times 1 = 4$

b. $2 \times 1 = 2$

c. 1

2. a.
$$\begin{bmatrix} 3-2i \\ 5 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3-2i+2 \\ 5+1 \end{bmatrix} = \begin{bmatrix} 5-2i \\ 6 \end{bmatrix}$$

b.
$$\begin{bmatrix} 2 \\ 7 \end{bmatrix} - \begin{bmatrix} 3+6i \\ 2i \end{bmatrix} = \begin{bmatrix} 2-(3+6i) \\ 7-2i \end{bmatrix} = \begin{bmatrix} -1-6i \\ 7-2i \end{bmatrix}$$

c.
$$\begin{bmatrix} 6 & 25 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 6+50 \end{bmatrix} = \begin{bmatrix} 56 \end{bmatrix}$$

d.
$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ 25 \end{bmatrix} = \begin{bmatrix} 6+50 \end{bmatrix} = \begin{bmatrix} 56 \end{bmatrix}$$

e.
$$\begin{bmatrix} 1 & 0 & 3 \\ 3 & 2 & 1 \\ 4 & 2 & 1 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 6 \\ 25 \end{bmatrix}_{2 \times 1} = \text{no multiplication possible}$$

because the columns of 1st matrix is not equal to the rows of the second matrix.

$$f. \begin{bmatrix} 1 & 0 & 3 \\ 3 & 2 & 1 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+0+3 \\ 3+0+1 \\ 4+0+1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 5 \end{bmatrix}$$

$$3. \begin{bmatrix} 3 \\ 4 \end{bmatrix} = x_0 |0\rangle + x_1 |1\rangle$$

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} = x_0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} x_0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ x_1 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} \quad x_0 = 3 \text{ and } x_1 = 4$$

$$4a. \langle 0|0\rangle = [1 \ 0] \begin{bmatrix} 1 \\ 0 \end{bmatrix} = [1+0] = [1]$$

$$b. \langle 0|1\rangle = [1 \ 0] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = [0+0] = [0]$$

$$c. |0\rangle\langle 1| = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$5a. A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad A^\dagger = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$AA^\dagger = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^\dagger A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$AA^\dagger = A^\dagger A = I$, the matrix is unitary.

$$b. A = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad A^\dagger = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$$

$$AA^\dagger = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$A^\dagger A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -i & 0 \end{bmatrix} \begin{bmatrix} i & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \end{bmatrix}$$

$AA^\dagger = A^\dagger A \neq I$, the matrix is not unitary

$$c. \quad A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad A^\dagger = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$AA^\dagger = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A^\dagger A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$AA^\dagger = A^\dagger A = I$, this matrix is unitary.

$$d. \quad \begin{bmatrix} 0 & -i \end{bmatrix} = A \quad A^\dagger = \begin{bmatrix} 0 & i \end{bmatrix}$$

$$\begin{bmatrix} i & 1 \end{bmatrix}$$

$$\begin{bmatrix} -i & 1 \end{bmatrix}$$

$$AA^{\dagger} = \begin{bmatrix} 0 & -i \\ i & 1 \end{bmatrix} \begin{bmatrix} 0 & i \\ -i & 1 \end{bmatrix} = \begin{bmatrix} -1 & -i \\ -i & 0 \end{bmatrix}$$

$AA^{\dagger}$ is not I , A is not an identity matrix.

$$b. a. \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

, is orthonormal.

$$b. \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 1/2 - 1/2 \end{bmatrix} = 0.$$

$$\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

are not orthonormal but are orthogonal.

$$c. \begin{bmatrix} 1/2 \\ 0 \\ 0 \\ 1/2 \end{bmatrix} = A \quad \begin{bmatrix} 1/2 \\ 0 \\ 0 \\ -1/2 \end{bmatrix} = B \quad \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = C \quad \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \\ 1 \end{bmatrix} = D$$

$$|A| = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

$$|B| = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

$$|C| = 1 + \frac{1}{2} = \frac{3}{2}$$

$$|D| = \frac{1}{2} + \frac{1}{2} + 1 = 2$$

$$|A| + |B| + |C| + |D| = \frac{1}{2} + \frac{1}{2} + \frac{3}{2} + 2 = \frac{1+1+3+4}{2} = \frac{9}{2}$$

$$A+B+C+D = \begin{bmatrix} \frac{1}{2} + \frac{1}{2} + 1 + \frac{1}{\sqrt{2}} \\ 0 \\ 1/\sqrt{2} \\ \frac{1}{\sqrt{2}} + 1 \end{bmatrix} = \begin{bmatrix} 2 + \frac{1}{\sqrt{2}} \\ 0 \\ 1/\sqrt{2} \\ \frac{1+\sqrt{2}}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{2\sqrt{2}+1}{\sqrt{2}} \\ 0 \\ 1/\sqrt{2} \\ \frac{1+\sqrt{2}}{\sqrt{2}} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} \frac{(2\sqrt{2}+1)\sqrt{2}}{2} \\ 2 \\ 0 \\ 1/\sqrt{2} \\ \frac{(1+\sqrt{2})\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} \frac{4+\sqrt{2}}{2} \\ 0 \\ 1/\sqrt{2} \\ \frac{\sqrt{2}+2}{2} \end{bmatrix} = F$$

$$|F| = \frac{16 + 8\sqrt{2} + 2}{4} + 0 + \frac{1}{2} + \frac{2 + 4\sqrt{2} + 4}{4}$$

$$= \frac{16 + 8\sqrt{2} + 2 + 2 + 2 + 4\sqrt{2} + 4}{4}$$

$$= \frac{28 + 12\sqrt{2}}{4} = \frac{13 + 6\sqrt{2}}{2}$$

$$7. |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

qubit state is given by

$$|\psi\rangle = \cos\theta |0\rangle + \sin\theta |1\rangle, \text{ given } \theta = 45^\circ$$

$$= \cos 45 |0\rangle + \sin 45 |1\rangle$$

$$= \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$$

Probability of measurement $|0\rangle$ is

$$P_1 = \cos^2 45 = \frac{1}{2} = 0.5$$

Probability of measurement $|1\rangle$ is

$$P_2 = \sin^2 45 = \frac{1}{2} = 0.5$$

$$8. \begin{bmatrix} \cos \theta \\ -\sin \theta \end{bmatrix} \begin{bmatrix} \sin \theta \\ \cos \theta \end{bmatrix}$$

$$\begin{bmatrix} \cos \theta \\ -\sin \theta \end{bmatrix} \cdot [\sin \theta \cos \theta]$$

$$= \cos \theta \sin \theta - \sin \theta \cos \theta = 0$$

$$|x_1| = \cos^2 \theta + \sin^2 \theta = 1$$

$$|x_2| = \sin^2 \theta + \cos^2 \theta = 1$$

So, new basis is orthonormal and
orthogonal